



Evaluating MAMBA on continuous sensor data

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Introduction

Mamba is a Transformer alternative that scales linearly with sequence length by using selective, input-dependent scanning. We investigate whether Mamba's advantage over Transformers, LSTMs, and CNNs grows as sequence length increases.

Methodology

Dataset: WESAD 15 subjects, 8 physiological channels (ACC (3), ECG, EDA, EMG, Resp, Temp), 3 affective states (baseline, stress, amusement).
Preprocessing: Downsampled 700 Hz → 100 Hz; Segmented into 60s, 90s and 120s sliding windows (30s stride).
Experimental Design: Subject-aware train/test split with class-balance loss; 70% – 15% – 15% (Train – Val – Test).
Models Compared: Mamba, Transformer, LSTM, and CNN trained under identical conditions.
Evaluation: Overall accuracy and F1 score, focus on stress detection performance.

Results

60s

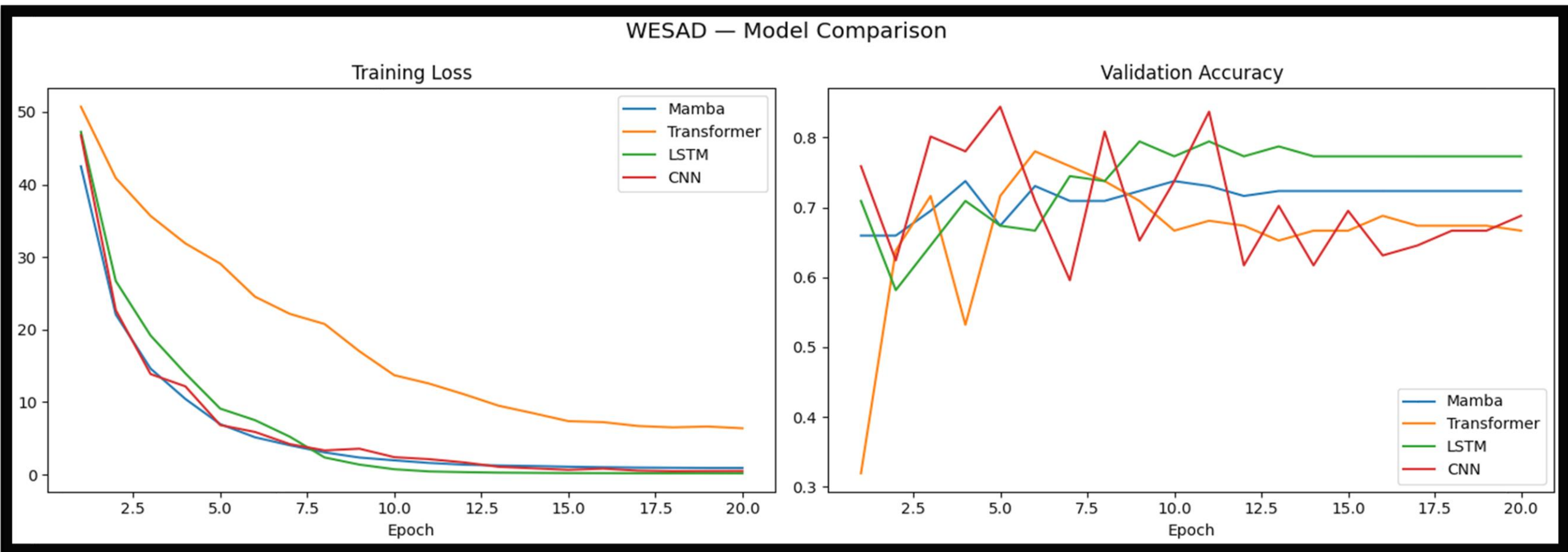


Figure 1. Training Loss and Validation Accuracy comparison between the four architectures using the 60s window size.

90s

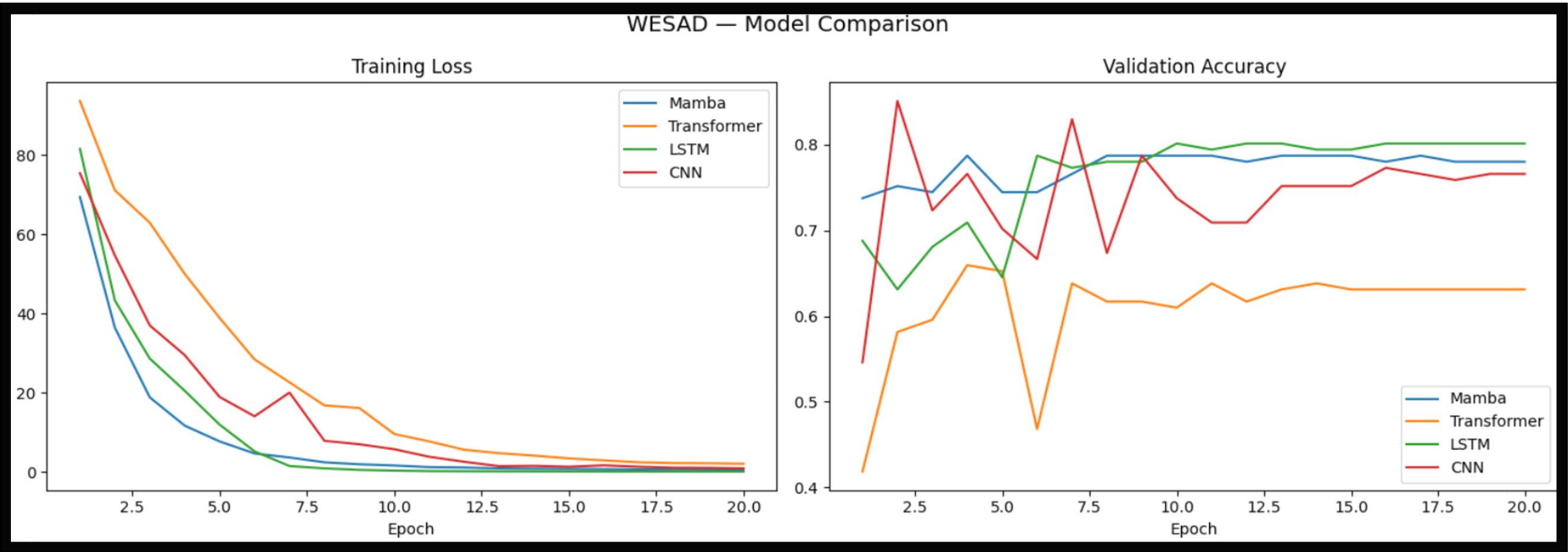


Figure 2. Training Loss and Validation Accuracy comparison between the four architectures using the 90s window size.

120s

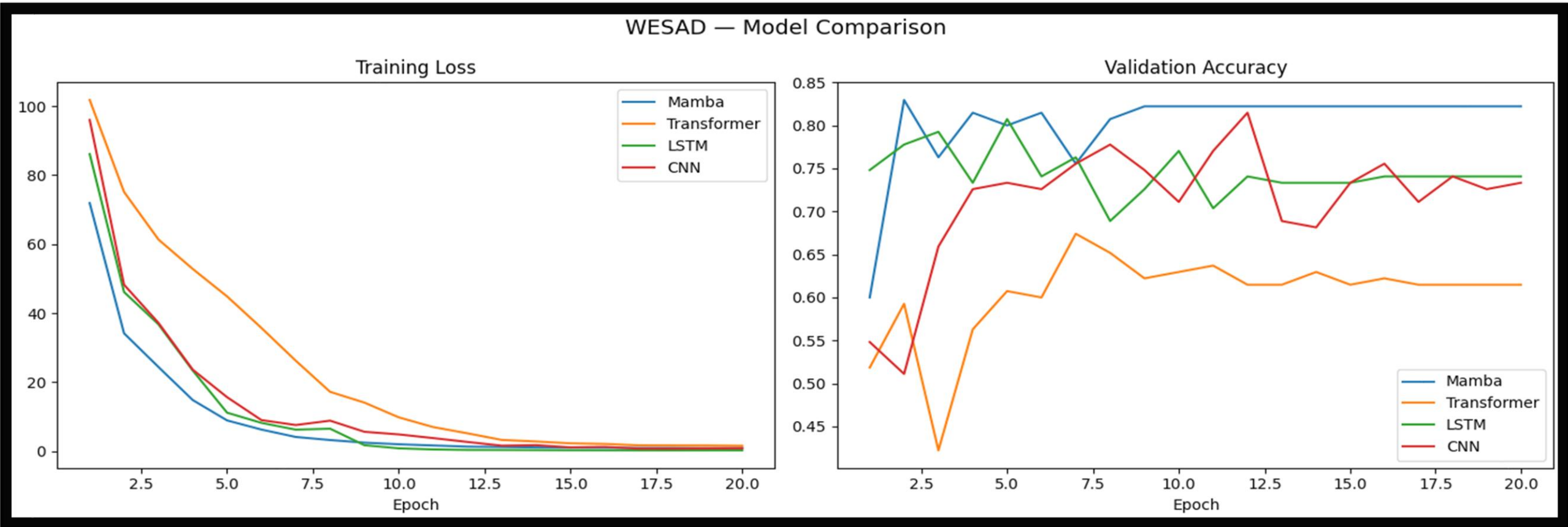


Figure 3. Training Loss and Validation Accuracy comparison between the four architectures using the 120s window size.

Architecture	Test Accuracy	F1 score
Mamba	72.86%	61.05%
Transformer	61.43%	34.41%
LSTM	65.24%	42.72%
CNN	72.86%	74.00%

Table 1. Test Accuracy and F1 score of the four architectures using the 60s window size.

Architecture	Test Accuracy	F1 score
Mamba	76.59%	67.29%
Transformer	41.95%	30.65%
LSTM	59.51%	36.19%
CNN	72.68%	74.23%

Table 2. Test Accuracy and F1 score of the four architectures using the 90s window size.

Architecture	Test Accuracy	F1 score
Mamba	68.34%	52.50%
Transformer	53.27%	55.17%
LSTM	36.68%	38.46%
CNN	74.87%	71.74%

Table 3. Test Accuracy and F1 score of the four architectures using the 120s window size.

Conclusion

- Mamba outperformed Transformer and LSTM across all window sizes
- Performance improved with longer context, supporting selective state spaces
- CNN remained highly competitive, achieving the best F1 score
- Local temporal patterns are strongly informative for stress detection
- Mamba offers the best overall balance of accuracy, stability, and scalability
- CNN's strong stress performance is worth further investigation

Future Work

- Deep Mamba architecture
- Bidirectional Mamba architecture
- Evaluating performance under simulated sensor dropout
- Subject covariate integration

References

Gu, Albert, and Tri Dao. "Mamba: Linear-time sequence modeling with selective state spaces." arXiv preprint arXiv:2312.00752 (2023).

Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger and Kristof Van Laerhoven, "Introducing WESAD, a multimodal dataset for Wearable Stress and Affect Detection", ICMI 2018.

